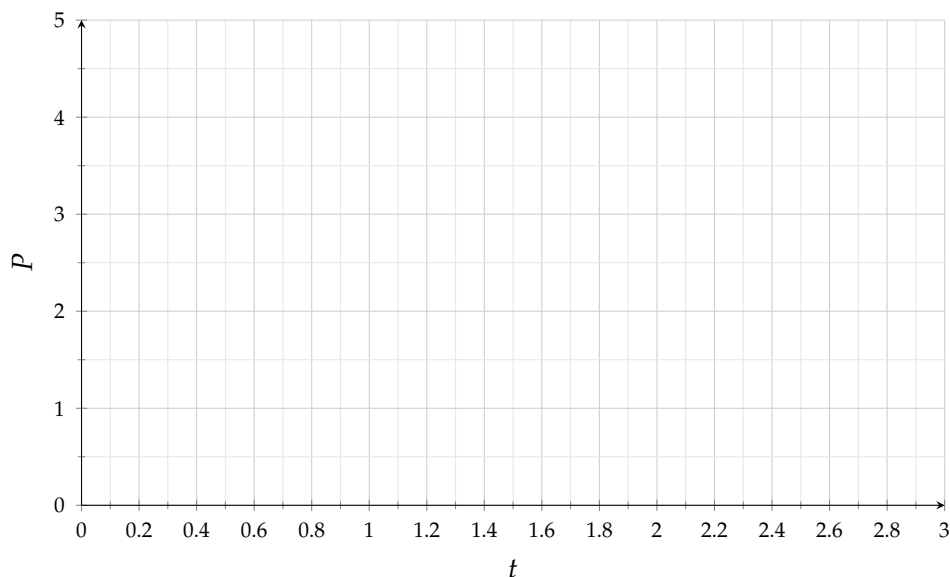


Lecture 10 Live Problems

The derivative. Exponential growth and the emergence of e .

Points toward lab manual pages: **Exp1, Exp2, Exp3, Exp4** (section L2 S3)

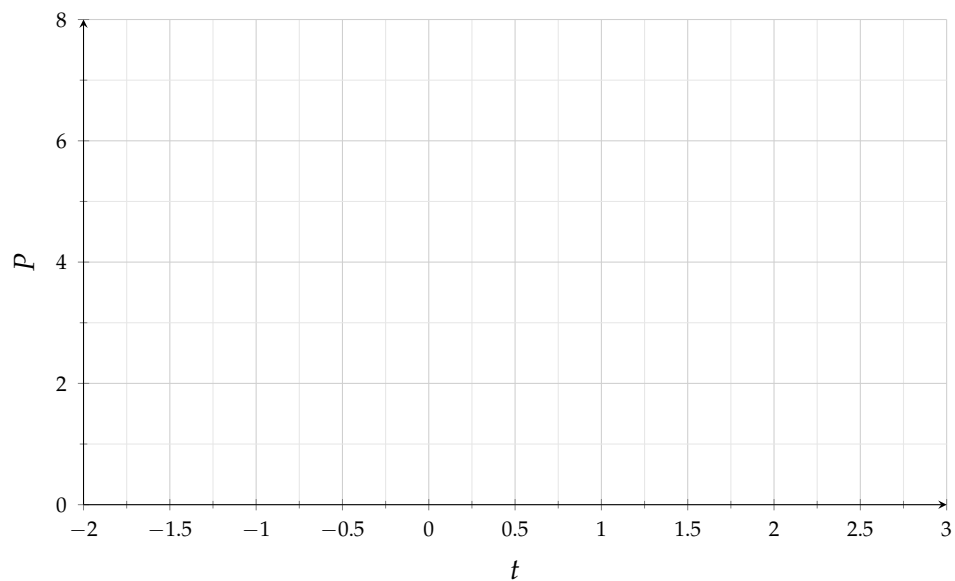
1. If $P(t) = t^2$, estimate dP/dt when $t = 1$. Sketch the graph of $P(t)$ and draw the tangent line at $t = 1$. Create a situation related to this value of dP/dt .



2. Examples of the power rule!

$A(t) =$	$dA/dt =$
$B(t) =$	$dB/dt =$
$C(t) =$	$dC/dt =$
$E(t) =$	$dE/dt =$
$F(t) =$	$dF/dt =$

3. If $P(t) = 2^t$, estimate dP/dt when $t = 0$. Sketch the graph of $P(t)$ and draw the tangent line at $t = 0$. Create a situation related to this value of dP/dt .



4. Four change equations. Which change equations are associated with linear growth? Exponential growth? What do their time-series plots look like? What situations do they describe? Which are discrete-time models and which are continuous-time models?

$$\frac{dP}{dt} = 3$$

$$\frac{dQ}{dt} = 3Q$$

$$\frac{\Delta R}{\Delta t} = 3$$

$$\frac{\Delta S}{\Delta t} = 3S$$